

URA302 | Pandas Basics – Complete Worksheet Solution

import pandas as pd

print("Q1. Pandas Basics")

1.1 Import pandas and check version

print("Pandas Version:", pd.__version__)

1.2 Create DataFrame

```
df = pd.DataFrame({
    "Name": ["Alice", "Bob", "Charlie"],
    "Age": [25, 30, 35],
    "City": ["New York", "Los Angeles", "Chicago"]
})
print("\nDataFrame:")
print(df)
```

print("\nQ2. Pandas Series")

2.1 Create Series

```
S1 = pd.Series([100, 200, 300, 400, 500])
print("Series S1:")
print(S1)
```

2.2 Access elements

```
print("Second element:", S1[1])
print("Fourth element:", S1[3])
```

2.3 Series addition

```
S2 = pd.Series([10, 20, 30, 40, 50])
print("S1 + S2:")
print(S1 + S2)
```

```
print("\nQ3. DataFrame Basics")

# 3.1 Column access
print("Name and City columns:")
print(df[["Name", "City"]])

# 3.2 Add Salary column
df["Salary"] = [50000, 60000, 70000]
print("\nUpdated DataFrame:")
print(df)

# 3.3 Statistics
print("Average Age:", df["Age"].mean())
print("Total Salary:", df["Salary"].sum())

print("\nQ4. Filtering and Indexing")

# 4.1 Conditional filtering
print("Age > 28:")
print(df[df["Age"] > 28])

# 4.2 Set and reset index
df_indexed = df.set_index("Name")
print("\nName as Index:")
print(df_indexed)

print("\nReset Index:")
print(df_indexed.reset_index())

print("\nQ5. Working with CSV Data")
# Creating CSV file (for demonstration)
employees = pd.DataFrame({
    "Name": ["John", "Jane", "Emily"],
```

```
    "Department": ["Sales", "Marketing", "HR"],
    "Salary": [50000, 60000, 55000]
})
employees.to_csv("employees.csv", index=False)
```

5.1 Read CSV

```
emp_df = pd.read_csv("employees.csv")
print("Employees DataFrame:")
print(emp_df)
```

5.2 CSV filtering

```
print("\nSalary > 55000 (Name & Department):")
print(emp_df[emp_df["Salary"] > 55000][["Name", "Department"]])
```

print("\nQ6. Grouping and Aggregation")

6.1 Average salary by department

```
print("Average Salary by Department:")
print(emp_df.groupby("Department")["Salary"].mean())
```

6.2 Min & Max salary

```
print("\nMin and Max Salary by Department:")
print(emp_df.groupby("Department")["Salary"].agg(["min", "max"]))
```

print("\nQ7. Merging DataFrames")

```
df1 = pd.DataFrame({
    "Name": ["John", "Jane", "Emily"],
    "Department": ["Sales", "Marketing", "HR"]
})
```

```
df2 = pd.DataFrame({
    "Name": ["John", "Jane", "Emily"],
```

```
"Experience (Years)": [5, 7, 3]
})
```

```
merged_df = pd.merge(df1, df2, on="Name")
print("Merged DataFrame:")
print(merged_df)
```

```
print("\nQ8. Sorting")
# 8.1 Sort by experience
sorted_df = merged_df.sort_values(by="Experience (Years)", ascending=False)
print("Sorted by Experience (Descending):")
print(sorted_df)
```